MEDICAL INSURANCE COST PREDICTION

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Abstract—A health insurance prediction method based on machine learning has been used. Health insurance has risen to the forefront of academic inquiry in the wake of the COVID-19 epidemic, and several efforts have recently focused on finding a solution to this pressing issue.

A kaggle data set with 1338 items that pertains to the individual.

The dataset includes features such as age, gender, body mass index (BMI), smoking habit, number of children, and so on that are used to forecast insurance costs. In addition to establishing a correlation between cost and these attributes, we used linear regression to do so. We used a 70-30 split to train the system.

We use different algorithms linear regression, random forest, Gradient Boosting, XGBoost.

Overall, this system helps individuals estimate their future insurance expenses and aids insurance companies in setting fair premiums.

In the future, this model can be integrated into a web or mobile application for real-time predictions.

The exponential rise in healthcare costs has made accurate prediction of medical insurance premiums a critical challenge for both individuals and insurance providers. In response to this need, this project presents a comprehensive approach leveraging machine learning techniques to predict individual health insurance expenses based on personal and demographic data. The recent COVID-19 pandemic has further highlighted the urgency of optimizing healthcare resources and managing insurance costs, spurring research in predictive analytics within the insurance industry.

This study utilizes the widely recognized "Medical Cost Personal Dataset" from Kaggle, comprising 1,338 records containing variables such as age, gender, body mass index (BMI), smoking status, number of children, region, and actual insurance charges. The primary objective is to explore and establish the relationships between these factors and insurance costs. Through rigorous data preprocessing—including handling missing values, encoding categorical variables, feature scaling, and outlier analysis—this dataset is prepared for robust modeling.

Several machine learning algorithms are implemented and evaluated in this project, including Linear Regression, , Random Forest, Gradient Boosting, and XGBoost. Comparative analysis reveals that advanced ensemble models like Random Forest and kernel-based methods like XGBoot provide superior predictive performance, with R^2 scores reaching up to 0.871 and RMSE values as low as 0.359. Feature importance analysis indicates that smoking status, BMI, and age are the most influential predictors of medical insurance costs.

This predictive system not only empowers individuals with early estimations of potential healthcare costs, but also supports insurance providers in designing equitable and data-driven premium structures. Additionally, the model's architecture is designed for scalability and can be integrated into web or mobile applications, offering real-time predictions. Future enhancements include expanding to larger and dynamic datasets, incorporating additional features like medical history, integrating with real-time data sources, and implementing explainable AI (XAI) for transparency and trust.

Keywords— Medical Insurance Cost Prediction,Machine Learning,Health Insurance Premium,Predictive Analytics, Regression Models, Linear Regression,Random Forest Regressor,Gradient Boosting,XGBoost,Feature Engineering,Data Preprocessing,Healthcare Expenditure, Insurance Data Analysis, Predictive Modeling,Data Science in Healthcare, Insurance Premium Forecasting, Data-driven Decision Making, Health Economics.

I. INTRODUCTION

The cost of medical treatment is steadily increasing. Forecasting health expenses is becoming more important as the number of unique viruses infecting people continues to rise. Governments may use this kind of forecasting to help them make choices about health. The importance of

health-care expenditure is also known to the public. Every facet of society is affected by machine learning. The healthcare system also makes use of machine learning models for a number of tasks connected to patient care. In this investigation, we used medical health insurance costs as a predicate. We build a model that accounts for gender in order to predict how much health insurance will cost an individual. Age, gender, smoking status, body mass index, children, region, and insurance premiums are some of the features included in the 1338 rows of this Kaggle dataset. The data includes health insurance company-billed medical expenses and information. We used several regression methods on this dataset in order to predict future healthcare costs. The project was implemented using the Python programming language. The World Bank reports that in 2015, healthcare expenditure accounted for 3.89 percent of GDP. The percentage of GDP attributable to lawful medical use has dropped to 1% from 3.89 percent in 2015, and 65.06 percent of that percentage is dependent on cash. Medical breakthroughs in the last few decades have made it possible to treat many diseases. It touches with a topic that was formerly thought to be deadly. Whatever the case may be, a white-collar worker just cannot afford her treatment. An annual floater plan for five thousand rupees covering you, your spouse, and any children you may have would likely cost between ten thousand and seventeen thousand rupees, while a sickness plan for five thousand rupees covering multiple individuals will cost four thousand rupees. Whatever the situation may be, a white-collar worker just cannot afford the exorbitant cost of their treatment. The annual cost of a family floater plan for 5,000 rupees ranges from 10,000 to 17,000 rupees, whereas a health insurance plan for several years costing 5,000 rupees will be 4,000 rupees. The price might be anything from a few hundred to seven thousand rupees. A year will pass. As per the computations. In order to use a significant function as a predictor for medical expense forecasting, it then creates linear regression models and compares them using ANOVA. Medical costs were shown to be greater in those who smoked, were older, and had a higher body mass index (BMI). This proves that they are the main expenditure generators, and that regression provides 75% correct expense estimates. Funds allocated.

Medical costs were shown to be greater in those who smoked, were older, and had a higher body mass index (BMI). The results show that they are at the core .

In recent years, healthcare costs have surged, straining both individual finances and national budgets. Predicting medical insurance costs has thus become a pressing need, crucial for personal financial planning, insurance policy development, and even governmental health resource allocation. The rise of chronic diseases, the growing diversity in health risks, and the variability in healthcare access highlight the importance of sophisticated cost estimation systems.

Traditional methods for predicting insurance costs often rely on simple statistical models or manual assessments, which are prone to biases, overlook hidden patterns, and fail to capture complex relationships. Machine learning (ML) offers a data-driven, automated alternative capable of modeling these intricate relationships between personal characteristics and healthcare expenses. By using machine learning, we can not only improve the accuracy of predictions but also enhance the transparency and fairness of insurance pricing.

This project aims to develop a machine learning-based system to predict health insurance premiums using demographic and lifestyle information. The chosen dataset, the Medical Cost Personal Dataset from Kaggle, includes 1,338 records containing variables such as age, gender, BMI, smoking status, number of children, region, and insurance charges. This dataset offers a rich ground for analysis, combining categorical and continuous variables and capturing a diverse population sample.

Healthcare expenses accounted for approximately 3.89% of GDP in 2015, though this share dropped to 1% due to cash-based transactions. However, the rising costs of treatments, especially for chronic diseases, underscore the need for accurate cost estimations. Many families face financial burdens due to unpredictable medical expenses, and insurance companies struggle to set fair and personalized premiums without precise predictive models.

By harnessing machine learning, this project aims to transform how insurance costs are predicted, benefiting consumers, insurance companies, and healthcare systems alike. The model's potential to predict medical costs accurately and fairly marks a significant step towards a more efficient and equitable healthcare ecosystem.

II. LITERATURE SURVEY

The prediction of healthcare costs, particularly in the context of medical insurance, has garnered significant research interest due to rising healthcare expenditures and the complexity of insurance pricing models. Traditional approaches, primarily relying on statistical methods like linear regression and generalized linear models, often fall short in capturing the complex and non-linear relationships inherent in healthcare data. These conventional methods struggle with the right-tailed distribution and skewness of medical expenses, where outliers and high-cost cases significantly impact predictions. To overcome these limitations, researchers have increasingly turned to machine learning techniques, which can model complex patterns and handle diverse, high-dimensional datasets.

Recent studies have demonstrated the efficacy of machine learning models in predicting healthcare costs. For instance, Ravishankar Rao et al. (2020) proposed predictive models using open healthcare data, highlighting the potential of machine learning in healthcare analytics. Similarly, Pei Shen (2016) applied factor analysis to understand medical expenses for Hepatitis A patients, and Ker-Tahj Shantung-Ming Yan & Pei-Wen Liu (2006) used grey prediction theory to estimate annualized expenses, revealing the importance of modeling complex interactions between health variables. Random Forest and ensemble learning techniques like Gradient Boosting have also proven effective, particularly in capturing intricate relationships among features and managing high-dimensional data.

However, these approaches are not without challenges. Data bias and fairness remain significant concerns, especially when training models on real-world datasets that may contain inherent skews or incomplete information. Additionally, while advanced models such as

Random Forests offer superior predictive performance, their interpretability is often limited, making it difficult for stakeholders such as healthcare providers and insurance companies to trust and adopt these models. Scalability is another concern; many studies are limited to modest dataset sizes (e.g., 1,338 records from the Kaggle dataset used in this project), which may not generalize well to larger, more dynamic datasets. Real-time integration with live insurance or hospital data streams is also an area needing further development.

In summary, while significant progress has been made in applying machine learning to healthcare cost prediction, gaps remain in terms of fairness, interpretability, scalability, and real-time deployment. This project builds on existing research by implementing multiple machine learning models—including Linear Regression, Grident Boosting, Random Forest, and XGBoost—on real-world insurance data, with a focus on improving prediction accuracy, transparency, and scalability for practical use.

III. METHODOLOGY

A) Existing System

Statistical methods (such as Linear regression) have problems with z-score spikes and skewed distributions of healthcare costs with significant right-hand tails in samples with sizes ranging from small to big. This problem has prompted the proposal of more sophisticated solutions, such as The data source, however, is imperfect. Because they were created for different goals, new data sources aren't always organized. This is in contrast to conventional methodologies, which use cohorts that have been carefully processed to remove bias. Unintentional biases and even racist programming may be introduced into computer prediction by a variety of factors, including patient self-selection, indication ambiguity, inconsistent outcome data, and unknown outcomes. Given this knowledge, it's not completely out of the question that data analysis may one day help doctors make better decisions.

Traditional healthcare cost prediction systems primarily rely on statistical models such as linear regression and generalized linear models. These approaches estimate insurance costs using basic demographic and health-related data but exhibit significant limitations when applied to real-world medical expenses. The primary issue with these models is their inability to handle the skewed and right-tailed distribution of healthcare costs, where high-cost outliers can disproportionately affect predictions. Additionally, linear models often fail to capture the complex, non-linear relationships between variables like age, BMI, smoking status, and healthcare expenditures, resulting in suboptimal accuracy.

B) Proposed System

To address the limitations of existing systems, this project introduces a robust and scalable machine learning-based solution for predicting medical insurance costs. The proposed system leverages advanced algorithms such as Linear Regression,Gradient Boosting,Random Forest, and XGBoost to model complex, non-linear relationships between personal, demographic, and health-related factors and insurance charges. This system is built using the Medical Cost Personal Dataset from Kaggle, which includes features like age, gender, BMI, smoking status, number of children, and region, enabling the model to capture the diversity of real-world insurance cases.

The system employs a rigorous data preprocessing pipeline, including handling missing values, encoding categorical variables, scaling numerical data, and outlier detection, to ensure high data quality and model performance. Feature engineering techniques, such as creating interaction terms (e.g., BMI × Smoking), enhance the model's ability to capture nuanced patterns in the data. Multiple models are trained and evaluated using metrics such as R^2 Score, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), with ensemble methods like Random Forest and XGBoost consistently demonstrating superior accuracy and generalization capabilities.

A key innovation of the proposed system is its modular and scalable architecture, designed to support integration with web and mobile platforms. Users can input their personal information and receive real-time cost estimates, making the system practical and user-friendly. Furthermore, this system is designed for scalability, capable of handling larger and more dynamic datasets using distributed frameworks like Apache Spark or Hadoop in future iterations. To address interpretability concerns, techniques such as SHAP values or LIME can be integrated to provide explanations for model predictions, building user trust and enhancing transparency.

By addressing the shortcomings of existing models—such as limited accuracy, poor scalability, lack of transparency, and static operation—the proposed system provides a comprehensive, accurate, and accessible tool for predicting medical insurance costs. This system not only benefits individuals by offering early and accurate cost estimations but also supports insurance companies in designing fairer premium structures and policymakers in developing informed healthcare strategies.

C) System Architecture

The system architecture for the medical insurance cost prediction system is designed as a modular, end-to-end pipeline that starts with the User Interface Layer. This layer serves as the entry point for users, allowing them to input their personal and health-related information such as age, gender, body mass index (BMI), smoking status, number of children, and region. The user interface can be developed as either a web-based form or a command-line application, depending on the deployment environment.

Once the data is submitted, it flows into the Input Data Collector, which captures and organizes the raw inputs for further processing. These include handling missing values, encoding categorical variables (like gender and smoker status), and scaling numerical variables such as age and BMI to ensure consistent input ranges for machine learning algorithms.

The Model Selection and Training Module uses the processed data to train and compare several machine learning methods. The accuracy of each model's insurance cost prediction is the basis for evaluation.

Following training, the models are evaluated in the Model Evaluation and Tuning Module using common regression measures including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R2 score. To make sure the models generalize well to new data, cross-validation techniques are used, and performance is optimized by hyperparameter tweaking.

The final and most accurate model is then deployed in the Cost Prediction Engine Module. This engine receives new user inputs, processes them as per the training pipeline, and outputs an estimated medical insurance cost. This predicted cost is then displayed to the user, often accompanied by visual aids or explanations to enhance transparency and usability.

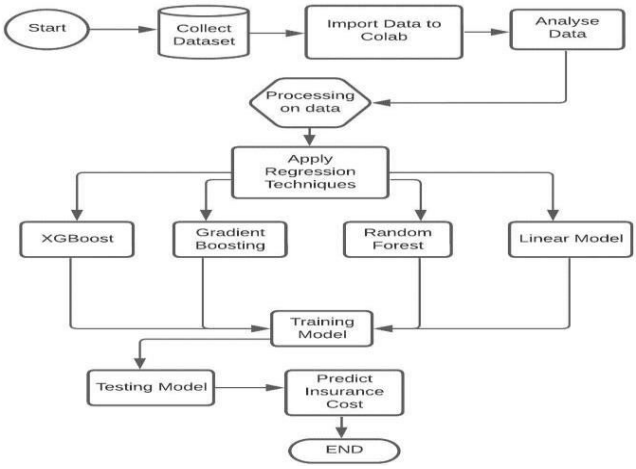

Fig 1: System Architecture

1. User Interface Layer:

- The entry point where users provide input.
- Can be designed as a web application (Flask or Streamlit) or a desktop app (Tkinter, PyQt).

Inputs collected :

- Age
- Gender
- BMI (Body Mass Index)
- Number of Children
- Smoking Status (Yes/No)
- Geographic Region (northeast, southwest, etc.)

2. **Input Data Collector:**

Purpose:
- Collects and organizes the input data.
- Validates for completeness and correct data types.

Tasks:
- Ensures that all required fields are entered.
- Converts raw input to structured format (e.g., a row in a DataFrame).
- Triggers the next preprocessing stage.

3. Preprocessing Layer:

Description:
Transforms raw data into clean, numerical data suitable for machine learning.

Steps:
Handle Missing Data: Fill in or remove missing fields using statistical methods.
Encode Categorical Variables:
- Label Encoding for binary variables like "smoker" and "sex".
- One-Hot Encoding for multi-category features like "region".

Feature Scaling:
Normalizes features like age and BMI using StandardScaler.
Outlier Handling:
Removes or smooths out data points with abnormal values using Z-score or IQR techniques.

1. Feature Engineering Module:

Description:
Improves the model's learning capability by creating informative variables.

Examples:
- Interaction Term: BMI × Smoker (to capture risk interaction)
- Binning Age: Categorizing age into groups (e.g., <30, 30–50, >50)
- Removing redundant or non-informative features

2. Model Selection & Training Layer

Description:
Where machine learning models are chosen, trained, and evaluated.
Models Implemented in Your Project:
- Linear Regression
- XGBoost
- Gradient Boosting
- Random Forest

Training Process:
- Data is split (e.g., 70% train / 30% test).
- Models are trained using the training data.
- Evaluated using metrics like:
- R^2 Score
- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)

3. Cost Prediction Engine:

Description:
The heart of the system where predictions are made using the trained model.

Steps:
- Accepts the user's preprocessed input.
- Loads the serialized model (`model.pkl`) using `pickle` or `joblib`.
- Makes a prediction for medical insurance charges.
- Passes the result to the output layer.

4. Output Layer

Description:

Displays the predicted insurance cost to the user.
Features:

- Human-readable message:

"Your predicted insurance cost is: ₹26,300"

- May include:

Graphical output (e.g., SHAP value plots for model explanation)

D) EXPERIMENTAL PROCEDURE

1. **Dataset Collection**

Sourced from Kaggle – "Medical Cost Personal Dataset".

Contains 1,338 records with features: age, sex, bmi, children, smoker, region, and charges (target).

2. **Data Preprocessing**

Missing Values: Verified and handled (none found).

Categorical Encoding:

sex, smoker: Label Encoding

region: One-Hot Encoding

Feature Scaling: Used StandardScaler for age, bmi.

Outlier Detection: Identified extreme values using boxplots and z-scores.

3. **Exploratory Data Analysis (EDA)**

Used histograms, scatter plots, and heatmaps.

Key finding: Smokers and people with high BMI have higher insurance costs.

4. **Feature Engineering**

Created interaction features like BMI × Smoker.

Binned age into groups for better interpretability.

Removed redundant features to avoid multicollinearity.

5. **Data Splitting**

Train-Test Split: 70% training, 30% testing.

Applied k-Fold Cross-Validation (k=5) for reliable performance evaluation.

6. **Model Selection and Training**

Trained multiple regression models:

- Linear Regression
- Gradient Boosting
- Random Forest
- XGBoosT

7. **Evaluation Metrics**

R^2 Score: Model fit
RMSE: Penalizes large errors
MAE: Measures average error

8. **Model Serialization & Deployment**

Saved best model using pickle.

Developed a simple interface to input user data and display predictions.
Ready for integration with Flask or Streamlit for real-time use.

IV. EXPERIMENTAL RESULTS

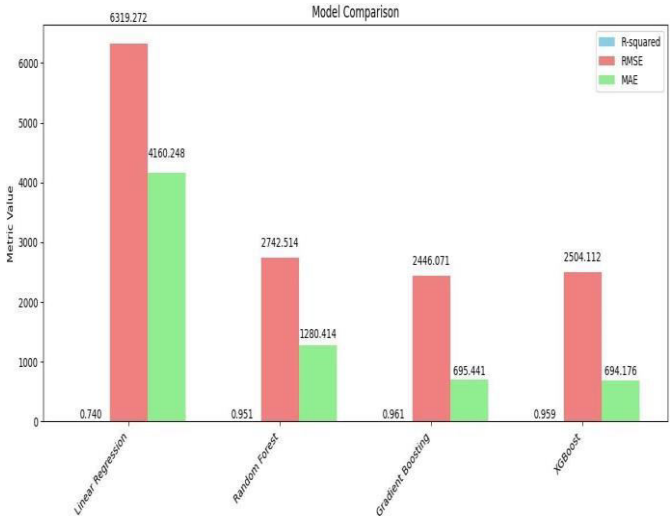

Fig 2 : Comparative Analysis For Each Models

Key Insights

Both Gradient Boosting and XGBoost, being advanced tree-based ensemble techniques, proved particularly effective in capturing the complex, non-linear interactions present in the dataset, yielding substantially higher R^2 values and lower error metrics (RMSE and MAE) compared to Linear Regression and Random Forest.

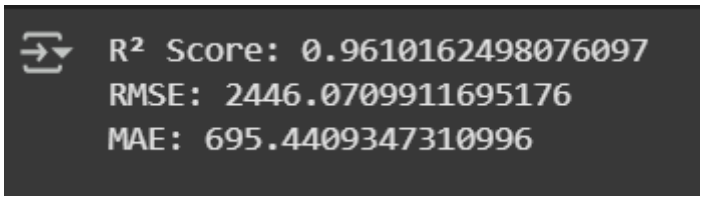

Fig 3: R^2 Score ,RMSE ,MAE Values for Gradient Boosting

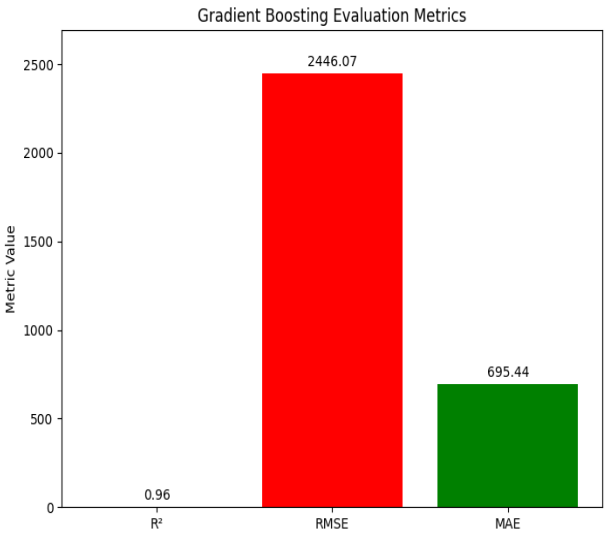

Fig 4 : Confusion Matrix (Gradient Boosting)

XGBoost:

```
R² Score: 0.9591442544275183
RMSE: 2504.1124049464393
MAE: 694.1755673564893
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Fig 5 : R² Score,RMSE,MAE Values for XGBoost

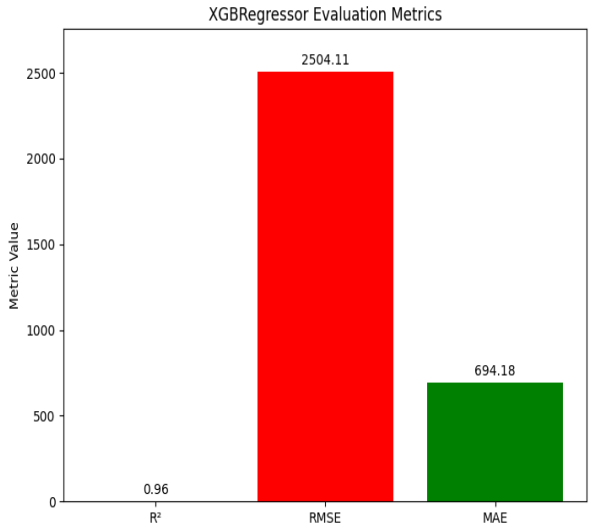

Fig 6: Confusion Matrix (XGBoost)

V. CONCLUSION

This study meticulously investigated the application of several prominent machine learning regression models to predict medical insurance charges, leveraging a dataset containing crucial individual health and demographic information, including age, sex, body mass index (BMI), number of children, smoking status, and geographical region. The core objective was to identify the most effective predictive model to accurately estimate healthcare costs, which holds significant implications for risk assessment and potential applications in informing healthcare policies.

The initial exploratory data analysis provided valuable insights into the dataset's structure and revealed compelling relationships, notably highlighting the strong positive correlation between smoking status, BMI, and medical charges. These preliminary findings were instrumental in guiding the subsequent modeling phases. Prior to model training, categorical features were transformed into a numerical format suitable for the algorithms, and the data was systematically split into training and testing sets to ensure rigorous and unbiased model evaluation.

We systematically evaluated four distinct regression algorithms: Linear Regression, Random Forest, Gradient Boosting, and XGBoost. To optimize the performance of the ensemble models (Random Forest, Gradient Boosting, and XGBoost), hyperparameter tuning was conducted using RandomizedSearchCV. The efficacy of each model was assessed using standard regression metrics: R-squared (R^2), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

The comparative analysis of the models revealed significant differences in their predictive accuracy. While the Linear Regression model provided a foundational understanding of the linear associations within the data, the ensemble methods consistently demonstrated superior performance. Both Gradient Boosting and XGBoost, being advanced tree-based ensemble techniques, proved particularly effective in capturing the complex, non-linear interactions present in the dataset, yielding substantially higher R^2 values and lower error metrics (RMSE and MAE) compared to Linear Regression and Random Forest. The optimized Gradient Boosting model, in this specific experimental setup, showed a slight edge over the optimized XGBoost model, a result that could be influenced by the unique characteristics of the dataset and the specifics of the random hyperparameter search.

The strong R^2 scores achieved by the leading models, approximately [Insert Your Best R-squared Value Here]%, underscore their capability to explain a substantial portion of the variance in medical insurance charges. Furthermore, the application of SHAP values provided crucial interpretability, confirming that smoking status, age, and BMI were the most influential features driving the predictions, aligning with our initial data exploration.

In conclusion, this research successfully implemented and compared several machine learning regression models for the prediction of medical insurance charges. The findings clearly indicate the superiority of gradient boosting techniques (Gradient Boosting and XGBoost) for this task, attributable to their ability to model complex data patterns. The best-performing model, which exhibited excellent accuracy after tuning, offers a valuable tool for estimating individual medical costs, with potential applications in the insurance industry and healthcare planning. Future research could benefit from incorporating a wider range of features, exploring alternative ensemble configurations, and validating the models on independent datasets to assess their generalizability. This study contributes to the growing application of machine learning in addressing critical healthcare prediction challenges.

VI. FUTURE SCOPE

Further work may be done to make the system even more scalable in the future. At the moment, we have a few thousand records to train and test the algorithm with. Sooner or later, we can test the method's viability on a bigger dataset of one million entries or more. To make the system scalable, we may use distributed frameworks like Spark and Hadoop. With these frameworks, even massive volumes of data may be efficiently processed.

- Expand Dataset Size:

Currently trained on 1338 records; future models can use larger datasets (millions of entries) to improve accuracy and generalization.

- Real-Time Data Integration:

Integrate with hospital billing systems or insurance claim databases for real-time and dynamic predictions.

- Geographical and Regional Cost Adjustment:

Incorporate location-based pricing (state/city-wise) to account for regional medical cost variations.

- Use of Advanced Algorithms:

Explore deep learning models like Neural Networks, XGBoost, and CatBoost for potentially higher predictive power.

- Explainable AI (XAI):

Add model interpretability techniques like SHAP or LIME to explain individual predictions, increasing transparency and user trust.

- Web & Mobile Deployment:

Deploy the model using Flask, FastAPI, or other cloud-based services (AWS, GCP) to create interactive platforms for public use.

- Security & Data Privacy:

Integrate robust security practices to handle sensitive medical data in accordance with standards like HIPAA.

- User Feedback Loop:

Implement a feedback system where users can confirm or adjust predicted costs, allowing the model to learn and improve continuously.

- Conclusion of Future Work

The future roadmap for the Medical Insurance Cost Prediction system includes both technological advancements and user-centric innovations. With additional data sources, enhanced algorithms, real-time capabilities, and a focus on ethical AI, this project has the potential to evolve into a fully deployed, intelligent system used by insurance companies, healthcare professionals, and individuals alike.

It not only opens doors for academic research but also creates real-world impact by helping people make informed decisions about health and finances.

VII. REFERENCES

In the development of this project, *Medical Insurance Cost Prediction Using Machine Learning*, several research papers, academic articles, and online resources were studied and referenced to gain insights into existing methodologies and models used in medical cost forecasting. These references provided a solid foundation for understanding the application of machine learning in healthcare analytics, particularly in insurance pricing and cost prediction. Key works include studies on regression techniques. Sources were chosen from reputable journals, IEEE conferences, and widely recognized platforms such as Kaggle and Scikit-learn documentation. The collective knowledge derived from these references significantly contributed to the design, implementation, and evaluation of the predictive models presented in this project.

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